

Q2 REPORT

Yusuf Eray Eriş

1. Introduction

In this project, I investigated Kovalevski configurations of smooth quartic curves over finite fields using Orbiter. A Kovalevski point is a point outside the quartic curve where four bitangents meet. For each given quartic curve, I computed the bitangents, the Kovalevski points, and the incidence structure between them.

The quartic curves were created in Orbiter using the normal form parameters given in the assignment. For each successfully computed case, Orbiter produced a LaTeX report containing the equation of the curve, the number of rational points, the 28 bitangents, the Kovalevski points, and the incidence matrix of the configuration.

The main object studied in this report is the incidence matrix between bitangents and Kovalevski points. In this matrix, the rows correspond to the 28 bitangents and the columns correspond to the Kovalevski points. An entry is equal to 1 if the corresponding bitangent is incident with the corresponding Kovalevski point, and 0 otherwise.

2. Orbiter Commands

The following Orbiter commands were used to construct the quartic curves and generate the reports.

```
Q2:
$(ORBITER) -v 3 \
  -define F9 -finite_field -q 9 -end \
  -define P9 -projective_space -n 2 -field F9 -v 0 -end \
  -define C9 -quartic_curve -space P9 -by_normal_form "6,3,4,6,2,5" -end\
  -define F11 -finite_field -q 11 -end \
  -define P11 -projective_space -n 2 -field F11 -v 0 -end \
  -define C11 -quartic_curve -space P11 -by_normal_form "8,5,3,2,10,7" -end\
  -define F17 -finite_field -q 17 -end \
  -define P17 -projective_space -n 2 -field F17 -v 0 -end \
  -define C17 -quartic_curve -space P17 -by_normal_form "2,7,16,3,9,4" -end\
  -with C9 -do \
    -quartic_curve_activity \
    -report \
  -end \
  -with C11 -do \
    -quartic_curve_activity \
    -report \
  -end \
  -with C17 -do \
    -quartic_curve_activity \
    -report \
  -end \
  -print_symbols
$(PDFLATEX) quartic_curve_by_normal_form_q9_report.tex
$(PDFLATEX) quartic_curve_by_normal_form_q11_report.tex
$(PDFLATEX) quartic_curve_by_normal_form_q17_report.tex
```

The activity `-report` was used as the main source of data. The command `-get_Kovalevski_configuration` was also tested, but for curves created directly by `-by_normal_form` it stopped with the message `QC->QOG == NULL`. Therefore, the incidence matrices were taken from the generated quartic curve reports. This is acceptable because the reports already contain the Kovalevski incidence structure and the matrix `Incma`.

3. Results

The following table summarizes the computed configurations.

Field	Parameters (a, b, c, d, e, f)	Points on curve	Bitangents	Kovalevski points	Singular points	Incidence matrix	Flags
GF(9)	(6, 3, 4, 6, 2, 5)	28	28	63	0	28×63	252
GF(11)	(8, 5, 3, 2, 10, 7)	0	28	21	0	28×21	84
GF(17)	(2, 7, 16, 3, 9, 4)	12	28	15	0	28×15	60

For the case over GF(9), Orbiter reports 28 points on the curve, 28 bitangents, 63 Kovalevski points, and no singular points. The incidence structure has 28 rows, 63 columns, and 252 flags.

For the case over GF(11), Orbiter reports 0 rational points on the curve, 28 bitangents, 21 Kovalevski points, and no singular points. The incidence matrix has size 28×21 with 84 flags.

For the case over GF(17), Orbiter reports 12 rational points on the curve, 28 bitangents, 15 Kovalevski points, and no singular points. The incidence matrix has size 28×15 with 60 flags.

Case GF(9)

For $q = 9$ and parameters (6,3,4,6,2,5)

Orbiter gives the quartic equation

$$\begin{aligned} X_0^4 + X_1^4 + 2X_2^4 + 4X_0^3X_1 + 8X_0^3X_2 + 6X_0X_1^3 + 6X_1^3X_2 \\ + 3X_0X_2^3 + 4X_1X_2^3. \end{aligned}$$

The curve has 28 rational points and 28 bitangents. Orbiter finds 63 Kovalevski points. The incidence matrix has size 28×63 and contains 252 flags. Since every Kovalevski point is incident with exactly four bitangents, the flag count agrees with $63 \cdot 4 = 252$.

This is the largest configuration among the computed cases.

Case GF(11)

For $q = 11$ and parameters $(8,5,3,2,10,7)$

Orbiter gives the quartic equation

$$\begin{aligned} X_0^4 &+ 4X_1^4 + 4X_2^4 + 5X_0^3X_1 + 4X_0^3X_2 + 4X_0X_1^3 + 4X_1^3X_2 \\ &+ 3X_0X_2^3 + 7X_1X_2^3 + 4X_0^2X_2^2 + 4X_1^2X_2^2 + 10X_0^2X_1X_2 \\ &+ X_0X_1^2X_2 + X_0X_1X_2^2. \end{aligned}$$

This curve has no rational points over GF(11), but it still has 28 bitangents and 21 Kovalevski points. The incidence matrix has size 28×21 and contains 84 flags. Again, the number of flags agrees with the fact that every Kovalevski point lies on four bitangents: $21 \cdot 4 = 84$.

Case GF(17)

For $q = 17$ and parameters $(2,7,16,3,9,4)$

Orbiter gives the quartic equation

$$\begin{aligned} X_0^4 &+ 4X_1^4 + 2X_2^4 + 15X_0^3X_1 + 6X_0^3X_2 + 6X_0X_1^3 + 14X_1^3X_2 \\ &+ 6X_0X_2^3 + 8X_1X_2^3 + 12X_0^2X_1^2 + 15X_0^2X_2^2 + 15X_1^2X_2^2 \\ &+ 3X_0^2X_1X_2 + 15X_0X_1^2X_2. \end{aligned}$$

This curve has 12 rational points, 28 bitangents, and 15 Kovalevski points. The incidence matrix has size 28×15 and contains 60 flags. This also satisfies $15 \cdot 4 = 60$. Compared with the cases $q = 9$ and $q = 11$, this case has fewer Kovalevski points.

The $q = 13$ Case

The assignment also lists the case $q = 13$, parameters = (3,2,10,4,2,0).

However, in my Orbiter build, this case could not be constructed from the normal form. Orbiter stopped during the construction of the quartic curve with the error

grassmann::rank_lint error, does not have full rank.

To check whether this was only caused by the last parameter $f = 0$, I tested the family

(3,2,10,4,2,f), $f \in \text{GF}(13)$.

In this test, the construction failed for all values of f except $f = 3$. Therefore, I did not include numerical configuration data for the assigned $q = 13$ case. This suggests that the assigned parameter set is problematic for the normal form construction in my Orbiter build. I therefore leave this case for further verification.

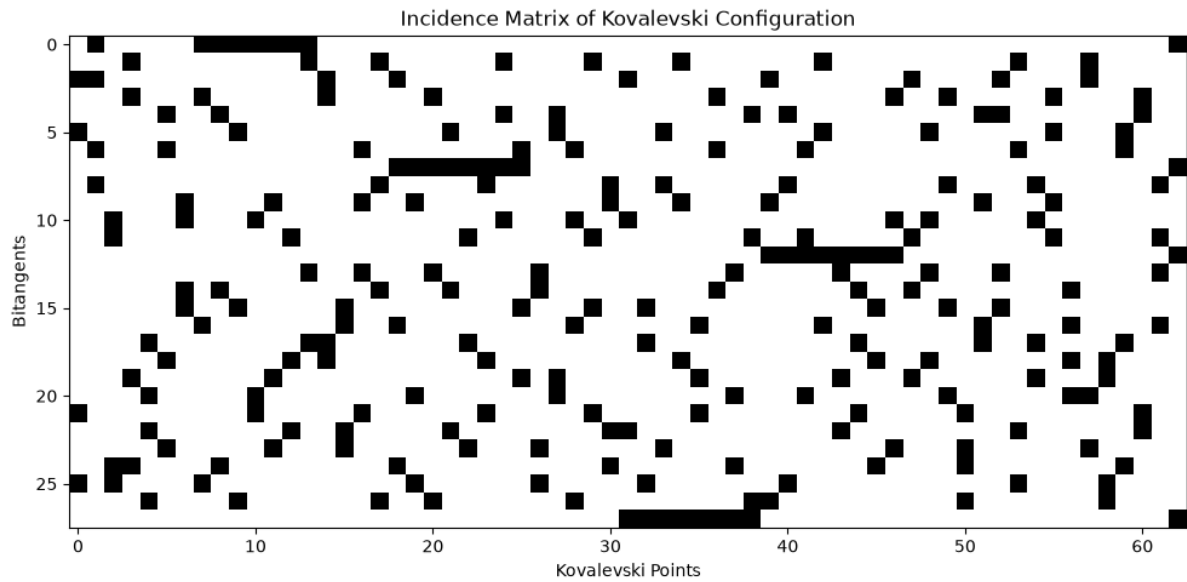
Incidence Matrix Bitmap Drawings

Orbiter prints the incidence matrix of the Kovalevski configuration in the generated report under the label `Incma`. In this matrix, the rows correspond to the bitangents and the columns correspond to the Kovalevski points. An entry equal to 1 means that the corresponding bitangent passes through the corresponding Kovalevski point. An entry equal to 0 means that there is no incidence.

For each computed case, I included the incidence matrix printed by Orbiter as a bitmap-style drawing. The black or filled entries represent incidences, while the empty entries represent non-incidences.

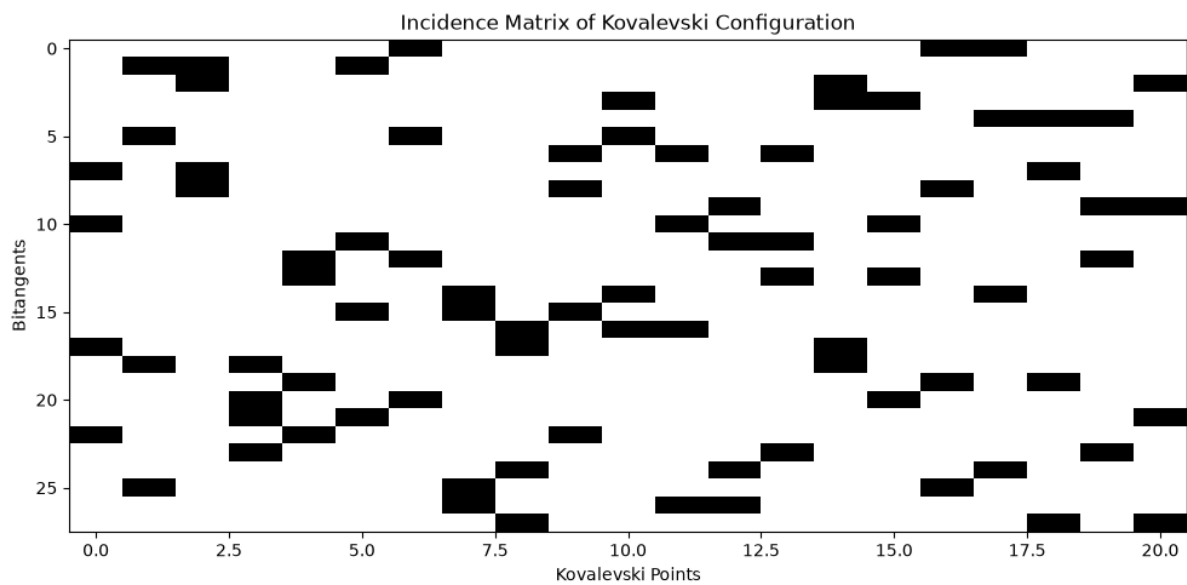
GF(9)

For the case over GF(9), the incidence matrix has size 28×63 with 252 flags.



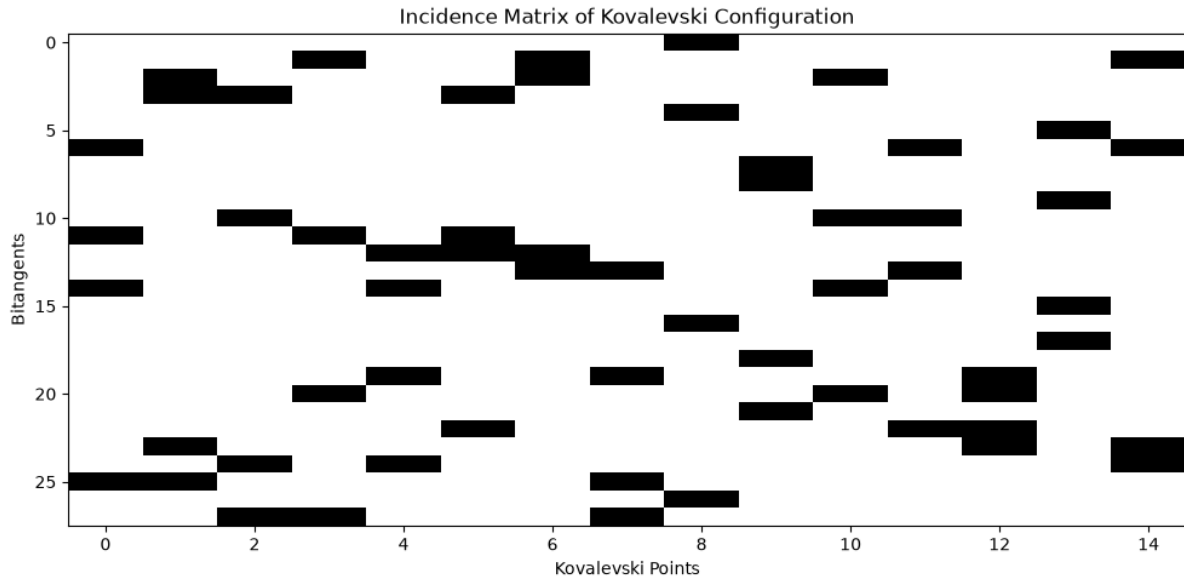
GF(11)

For the case over GF(11), the incidence matrix has size 28×21 with 84 flags.



GF(17)

For the case over GF(17), the incidence matrix has size 28×15 with 60 flags.



Conclusion

The computed cases show that all successfully constructed curves are nonsingular and have 28 bitangents. However, the number of Kovalevski points depends strongly on the field and the parameters. The $q = 9$ case gives the largest configuration, with 63 Kovalevski points and 252 incidences. The $q = 11$ and $q = 17$ cases give smaller configurations, with 21 and 15 Kovalevski points respectively.

In each successful case, every Kovalevski point is incident with exactly four bitangents. This is confirmed by the flag counts:

$$\text{GF}(9): 63 \cdot 4 = 252$$

$$\text{GF}(11): 21 \cdot 4 = 84$$

$$\text{GF}(17): 15 \cdot 4 = 60$$

The $q = 13$ case could not be completed with the assigned parameters because Orbiter failed during the normal form construction. This case requires either further investigation or confirmation of the parameter list.